



IILM University Greater Noida
School of Computer Science and Engineering

B. Tech. (CSE)

Course Booklet

(2026-2030)

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Abbreviations

PEO	Programme Educational Objectives
PO	Programme Outcomes
VAC	Value Added Courses (VAC)
MDC	Multidisciplinary Courses (MDC)
AEC	Ability Enhancement Courses (AEC)
SEC	Skill Enhancement Courses (SEC)
PE	Professional Electives

About the School

School of Computer Science & Engineering is one of the leading school at IILM University that runs the B. Tech in Computer Science & Engineering, specialization in AI/ML, Data Science & Big Data Analytics, Cloud Computing and Cyber Security domains, M.Tech., and Ph.D. programmes. In recent years, the significance of computing education has grown significantly, and our school stands as a leader, committed to shaping future technology pioneers. Our primary objective is to empower and guide students to become industry leaders in cutting edge computing technologies on both national and global scales. Our esteemed faculty, with extensive experience, is dedicated to providing students with a unique blend of theoretical knowledge and practical skills necessary for success in the ever-changing fields of Computer Science.

From innovative research to practical projects, we support our students in navigating the evolving landscape of computer science, encompassing areas such as Artificial intelligence, Machine learning, Cloud computing, Cyber security, Digital forensics, data analysis, Big data technologies, Data visualization, Generative-AI and Business intelligence. Our curriculum is designed to ensure students acquire a comprehensive understanding of the latest technologies and methodologies driving innovation in this data-driven era.

Aligned with the NEP2020 framework, the curriculum at the School of Computer Science & Engineering features a student-centric flexible structure, multidisciplinary learning, compulsory ability enhancement and skill enhancement courses, value-added courses, product design and development, as well as compulsory internships with multiple exit and entry points. Additionally, we employ innovative teaching and learning methodologies such as Outcome-Based Education (OBE), Project-Based Learning (PBL), Interleaving (Context Switching), and Pair Programming to enhance the learning experience.

Vision and Mission

Vision:

To lead globally in computer science and engineering education and research, driving innovation and societal impact.

Mission:

M1: Provide globally relevant education in computer science and engineering.

M2: Cultivate a culture of inclusivity, diversity, and excellence.

M3: Conduct cutting-edge research to address global challenges.

M4: Foster ethical leadership and responsible innovation in technology.

M5: Promote sustainable development and positive societal impact through technology.

Programme Educational Objectives

PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To cultivate strong analytical and computational thinking capabilities enabling graduates to model, interpret, and solve AI-driven problems using mathematical rigor, statistical reasoning, and engineering foundations.

PEO2: To develop proficiency in applying algorithms, machine learning, deep learning, data engineering, and cloud-based intelligent systems to design and deploy scalable AI solutions across diverse domains.

PEO3: To foster innovation, collaborative learning, leadership, and teamwork through interdisciplinary AI projects, research initiatives, entrepreneurial activities, and industry-academia engagements.

PEO4: To cultivate integrity, transparency, and ethical reasoning in AI development, ensuring responsible data use and fairness, and enabling graduates to advance AI that supports sustainable social progress.

PROGRAM OUTCOMES (POs) and PROGRAM SPECIFIC OUTCOMES (PSO)

1. PROGRAM OUTCOMES (POs)

After completion of the program, the students will be able to:

PO1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3. Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to

one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

B. Program Specific Outcomes (PSOs): for B. Tech in Artificial Intelligence

PSO1: Apply the foundations of computing, mathematics, statistical reasoning, and algorithmic thinking to analyze data, model complex problems, and develop intelligent, data-driven solutions using appropriate artificial intelligence and machine learning approaches.

PSO2: Design, implement, optimize, and deploy AI-based systems by integrating software engineering principles, modern development methodologies, cloud computing, MLOps, and model governance practices to solve real-world challenges effectively and efficiently.

PSO3: Utilize a diverse range of AI methodologies including natural language processing, computer vision, reinforcement learning, generative AI, and responsible AI frameworks to build scalable, ethically aligned, and human-centered intelligent systems across multidisciplinary domains.

B.Tech. AI (2025-2026)							
Semester 1:							
Sl. No.	Course Type	Course Code	Course Name	Credits	L	T	P
1	Multidisciplinary	CSE26101	Semiconductor and Quantum Physics*	3	3	0	0
2	Major Core	CSE26102	Digital Electronics and Computer Organization*	4	3	1	0
3	Multidisciplinary	CSE26103	Applied Calculus	3	2	1	0
4	Major Core	CSE26104	Computational Design Thinking	2	1	1	0
5	AEC	CSE26105	Professional Communication for Engineers*	2	2	0	0
6	VAC	CSE26106	Environmental Studies & Disaster Management (MOOC)*	2	2	0	0
7	Major Core	CSE26107	Foundation of AI and Automation	2	2	0	0
8	Major Core	CSE26108	Programming in C	3	3	0	0
9	Multidisciplinary	CSE26101P	Semiconductor and Quantum Physics Lab*	1	0	0	2
10	Multidisciplinary	CSE26103P	Applied Calculus lab	1	0	0	2
11	Major Core	CSE26108P	Programming in C Lab	2	0	0	2
12	SEC	CSE26111	Linux Administration Lab	1	0	0	2
13	Major Core	CSE26102P	Digital Electronics and Computer Organization Lab*	1	0	0	2
14	SEC	CSE26109	Entrepreneurial Mindset	2	2	0	0
Total				20	14	3/2	6/8

Semester 1

IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year	I	Course Name	<i>Semiconductor and Quantum Physics</i>	
	Program :	B.Tech. (CSE)	Semester	I	Course Code	CSE26101	
					L-T-P-C	3-0-0-3	
					Prerequisite Subject	Intermediate Mathematics and Physics	
Course Objectives		1. Understand and Apply Quantum Mechanics Fundamentals.					
		2. Analyze Quantum States and Measurement Processes.					
		3. Explore the Physical Realization of Quantum Systems.					
		4. Fundamentals of Semiconductor Physics and Devices.					
Course Outcome (At the end of the course, students will be able to)		Description					
CO1		Understand the foundational principles of classical mechanics and their limitations					
CO2		Develop a strong foundation in the principles of quantum mechanics.					
CO3		Understand the mathematical tools and concepts essential for quantum computing.					
CO4		Develop a strong foundation in semiconductors and superconductors					
Course Description		This course is designed to provide students with a comprehensive understanding of the fundamental principles of quantum mechanics as they apply to quantum computing. The focus will be on developing the mathematical tools and concepts necessary for the analysis and manipulation of quantum systems, with a specific emphasis on their relevance to quantum computing. The course will cover topics such as matrix representation, spin matrices, photon states, density matrices, quantum measurement, and operators. Second part of the course is focus on the semiconductor devices and its application.					
Detailed Syllabus							
Unit	Topics					CO No.	Number of lectures
Unit 1: Foundation of classical Physics and and Vector Analysis	Overview of the physics and its development, Classical mechanics and its limitations, Degree of freedom and its significance in real life phenomena, co-ordinate system, cartesian, spherical, cylindrical, curl gradient, divergence and its applications (cartesian, spherical, cylindrical). Equation of motions, problem solving. Electrostatics: Columbs law, Gauss law and its application, boundary condition of electrostatics, polarization, Maxwell's law.					CO1	10
Unit 2: Origin of Quantum Theory	The limitations to existing theories of black body radiation. Planck's hypothesis and its successes. Einstein explanation of the photoelectric effect. Wave particle duality, Uncertainty Principle, Wave function, Properties of Wave function, Interpretation of wave functions. Probability of current densities and Normalization.					CO2	10

Unit 3: Wave Function and Schrodinger Wave Equation	Time independent Schrodinger equation and its application. Energy Eigen values and Eigen functions. Position momentum and energy operators. Time dependent Schrodinger equation and dynamical evolution of the quantum state; Commutator of position and momentum operators; Expectation values of position and momentum. Wave function of a free particle. Postulates of Quantum mechanics. Vector spaces and subspaces, State vectors and matrix representation, Bra-ket notation and Dirac notation, Inner product spaces and orthogonality, Unitary and Hermitian matrices. Introduction to spin-1/2 systems and their properties, Spin operators and their mathematical representation, Quantum spin and Pauli matrices, Spin eigenstates and the Bloch sphere representation, Applications of Pauli matrices in quantum gates .	CO3	14											
Unit 4: Introduction to Semiconductor Physics	Crystal structure of Semiconductors, Energy Bands, Electrons and holes, Fermi Dirac distribution, Fermi Level, Intrinsic carrier concentration, Extrinsic Semiconductors, PN Junction Diode: Characteristics and application, Bipolar Junction Transistor: Characteristics and application. MEISSNER EFFECT, TYPE I and Type II superconductors	CO4	10											
Text Books :														
1. Zettili, N., 2009. Quantum mechanics: concepts and applications. John Wiley & Sons														
2. Neamen, D., 2005. <i>An introduction to semiconductor devices</i> . McGraw-Hill														
Reference Books :														
1. Introduction to Quantum Mechanics by David J. Griffiths."														
2. "Quantum Mechanics: Concepts and Applications" by Tarun Biswas														
3. Quantum computation and quantum information by Nielsen, Michael A., and Isaac L. Chuang.														
Online Resources:														
1. “Quantum Mechanics” by Prof. R. Shankar from Yale University , Youtube link: https://www.youtube.com/watch?v=uK2eFv7ne_Q&list=PLozLiCENL19jyrZvpUvG3W79bHKUOY79m														
Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination														
Examination Scheme:														
Components		MSE	Presentation/Assignment/ etc	ESE										
Weightage (%)		20%	30%	50%										
Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)														
CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2	2	1	2	–	–	–	–	–	–	2	3	3
CO2	3	3	3	2	2	–	–	–	1	–	–	2	2	3
CO3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO4	3	2	2	2	2	–	–	–	–	–	–	2	3	3
Average	3	2.4	2.4	2	2.2	0.6	0.6	0.6	0.8	0.6	0.6	2.2	2.8	3

 IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name:	Semiconductor and Quantum Physics Lab
	Program:	B. Tech. (CSE)	Semester:	I	Course Code:	CSE26101P
					L-T-P-C	0-0-2-1
					Prerequisite Subject:	Digital Electronics
Course Objectives		1. Understand the theoretical foundation of electromagnetic wave formation and conduction				
		2. Explain the dual nature of matter using experiments such as the Compton effect and photoelectric effect.				
		3. Evaluate the Principles and Applications of Lasers, Optical Fibers, and Semiconductor Materials				
		4. Analyze quantum concepts essential for modern physics and engineering applications				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		Applications of Maxwell’s equations to understand the formation and conduction of electromagnetic wave in different medium.				
CO2		To understand the dual nature of matter with the explanation of Compton effect and Schrodinger wave equation.				
CO3		Identify the energy band in solids and electron occupation probability to understand the physics of semiconductors and superconductors.				
Course Description		This course provides a foundational understanding of electromagnetic theory, quantum mechanics, and solid-state physics for engineering applications. It covers the application of Maxwell’s equations to electromagnetic wave propagation, the dual nature of matter through quantum phenomena like the Compton effect and Schrödinger equation and explores the working principles of lasers and optical fibers. The course also introduces energy band theory to explain the behavior of semiconductors and superconductors in modern electronic devices.				
List Of Experiments	1. Determination of Bandgap of Semiconductor – Four Probe Method To determine the energy bandgap of a semiconductor using the four-probe method.					
	2. Determination of Planck Constant using LED Method To calculate Planck’s constant by studying the voltage–wavelength relationship of LEDs.					
	3. Determination of Beam Divergence of LASER To measure the beam divergence of a laser using geometric methods.					
	4. I-V Characteristics of p-n Junction Diode To study the current-voltage characteristics of a p-n junction diode.					
	5. Characteristics of Zener Diode and Voltage Regulation To study the Zener diode characteristics and observe its role in voltage regulation.					
	6. Study of Half and Full Wave Rectifier To analyze the performance of half-wave and full-wave rectifiers.					
	7. Characteristics of Common Emitter NPN/PNP Transistor To study the input and output characteristics of a common emitter transistor and determine current and voltage gain.					
	Experiment 1:	Title: Determination of Bandgap of Semiconductor – Four Probe Method Objective: To determine the bandgap of a semiconductor using the four-probe method.				
Lab Activities: • Set up the four-probe apparatus. • Record resistance at different temperatures. • Plot graphs to calculate bandgap energy.						

Experiment 2:	Title: Determination of Planck Constant using LED Method Objective: To calculate Planck's constant using the LED voltage–wavelength relation. Lab Activities: • Connect different LEDs in circuit. • Measure threshold voltage for each color. • Plot energy vs. frequency to calculate Planck's constant.	
Experiment 3:	Title: Determination of Beam Divergence of LASER Objective: To measure the divergence angle of a laser beam. Lab Activities: • Place laser at fixed position. • Measure spot size at different distances. • Calculate divergence using geometry.	
Experiment 4:	Title: I-V Characteristics of p-n Junction Diode Objective: To study forward and reverse bias characteristics of a diode. Lab Activities: • Connect diode in forward bias and record readings. • Connect diode in reverse bias and record readings. • Plot I-V graphs for both conditions.	
Experiment 5:	Title: Characteristics of Zener Diode and Voltage Regulation Objective: To analyze Zener diode characteristics and study its voltage regulation property. Lab Activities: • Connect Zener diode in reverse bias. • Record voltage and current values. • Study regulation by varying load resistance.	
Experiment 6:	Title: Study of Half and Full Wave Rectifier Objective: To study the rectification process in half and full wave rectifiers. Lab Activities: • Connect diode in half-wave rectifier circuit. • Record output waveform on CRO. • Repeat with full-wave rectifier and compare efficiency.	
Experiment 7:	Title: Characteristics of Common Emitter NPN/PNP Transistor Objective: To study transistor input/output characteristics and calculate current and voltage gain. Lab Activities: • Connect transistor in CE configuration. • Measure input and output characteristics. • Calculate β (current gain) and voltage gain.	
Text Books:		
1. S.P. Taneja, Modern Physics for Engineers, 7th edition, 2010, R Chand And Company, India 2. S. K. Gupta, Engineering Physics I and II, 2012.		
Reference Books :		
1. D. J. Griffith, Introduction to Electrodynamics, 4th edition, 2015. 2. Arthur Beiser, Concepts of Modern Physics, 7th edition, 2017.		
Online Resources:		

**Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination
Examination Scheme:**

Components	Continuous Assessment	Lab Test / Presentation / Assignment / etc.
Weightage (%)	50%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2	1	-	-	-	-	-	-	-	-	2	3	-
CO2	3	3	2	-	2	-	-	-	1	-	-	2	3	2
CO3	3	3	3	3	2	2	-	-	-	2	-	3	3	3
CO4	3	2	2	2	2	2	-	-	-	-	-	2	2	3
Average	3	2.4	2	1.4	1.4	1.4	-	-	0.2	0.4	-	4.6	2.6	2.2

IILM University, Greater Noida	School:	School of Computer Science & Engineerin g	Year	I	Course Name	<i>Digital Electronics & Computer organization</i>
					Course Code	CSE26102
	Program:	B.Tech. (CSE)	Semester	I	L-T-P-C	3-0-0-3
					Prerequisite Subject	Semiconductor Devices
Course Objectives		5. Understand and apply number systems and Boolean algebra in digital circuit design.				
		6. Design and implement combinational and sequential logic circuits.				
		7. Analyze synchronous sequential circuits including counters and registers.				
		8. Understand the internal organization of a computer system including CPU, memory, and buses.				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		Convert and analyze different number systems, binary arithmetic, Boolean expressions, and implement them using logic gates.				
CO2		Design combinational logic circuits using SOP/POS forms and minimization techniques including multiplexers, encoders, and decoders.				
CO3		Design synchronous sequential circuits using flip-flops, counters, and shift registers.				
CO4		Explain the functional components of a computer system including the CPU, ALU, registers, buses, and Von Neumann architecture.				
CO5		Describe memory organization, memory hierarchy, instruction execution cycle, and CPU operation.				
Course Description		This course provides a comprehensive introduction to digital electronics and computer organization. It covers number systems, binary arithmetic, Boolean algebra, and logic gate implementation, enabling students to design and simplify combinational and sequential logic circuits. The course also explores key components of computer architecture, including the control unit, ALU, memory hierarchy, and instruction cycle, based on the Von Neumann model. By bridging digital logic design with the structure and operation of computer systems, students gain foundational knowledge essential for understanding hardware functionality and system-level behaviour in modern computing.				
Detailed Syllabus						
Unit	Topics			CO No.	Number of lectures	

Unit 1: Number Systems and Codes	Review of number systems Binary, Octal, Decimal, Hexadecimal and their interconversions, Binary Arithmetic, Signed Binary Numbers, Error detection, Boolean Algebra Laws and theorems, De Morgan's theorem algebra & simplification of Boolean expressions, Implementation of Boolean expressions, Logic Gates: Basic (AND, OR, NOT), Universal (NAND, NOR), XOR, XNOR	CO1	10
Unit 2: Combinational Logic Design	Combinational circuit designing, SOP, POS forms Minimization Techniques of Boolean functions such as Karnaugh map, multiplexers, demultiplexers, encoders, decoders & comparators.	CO2	10
Unit 3: Synchronous Sequential Circuit	Introduction to Sequential Logic: Need for sequential circuits, Binary cell, Latches and different types of Flip-flops, Master Slave Flip-flops. Edge Triggered Flip-flops. Fundamentals of Synchronous sequential circuits, Clocked Sequential Circuits, Design of Synchronous and Asynchronous Counters, Shift registers & Ring Counters	CO3	8
Unit 4: Structure of Computers	Computer types, Functional Units of a Computer: Control Unit, ALU, Registers, Buses, basic operational concepts, Von-Neumann architecture, bus structures, software, performance, multiprocessors and multicomputer	CO4	8
Unit 5: Processor and Memory	Registers, Address Space, Memory Locations, Memory Operations, Memory Hierarchy: Cache, Main Memory, Secondary Storage, CPU Organization: Instruction cycle, Fetch-decode-execute	CO5	8
Text Books :			
1. M. Morris Mano and Michael D. Ciletti , <i>Digital Design</i> , 5th Edition, Pearson Education, 2013. 2. Carl Hamacher, Zvonko Vranesic, and Safwat Zaky , <i>Computer Organization and Embedded Systems</i> , 6th Edition, McGraw-Hill, 2012.			
Reference Books :			
1. William Stallings , <i>Computer Organization and Architecture: Designing for Performance</i> , 11th Edition, Pearson Education, 2020 2. Ramesh S. Gaonkar , <i>Microprocessor Architecture, Programming and Applications with the 8085</i> , 6th Edition, Penram International, 2013. 3. Thomas L. Floyd , <i>Digital Fundamentals</i> , 11th Edition, Pearson Education, 2015. 4. John P. Hayes , <i>Computer Architecture and Organization</i> , 3rd Edition, McGraw-Hill, 1998.			
Online Resources:			
1. Introduction to Digital Computer Organization by Prof. P.K. Biswas A foundational YouTube lecture covering CPU internals from IIT Kharagpur https://www.youtube.com/watch?v=TH9nd-KdVHs 2. Digital Electronic Circuits IIT Kharagpur (Swayam) https://onlinecourses.nptel.ac.in/noc20_ee32/preview?			

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination
Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	PO 1	P O2	P O3	PO 4	PO 5	P O6	PO 7	PO 8	PO 9	P O 10	P O 11	P O 12	P S O 1	P S O 2
CO1	3	2	2	1	2	–	–	–	–	–	–	2	3	3
CO2	3	3	3	2	2	–	–	–	1	–	–	2	2	3
CO3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
CO4	3	2	2	2	2	–	–	–	–	–	–	2	3	3
CO5	3	2	2	2	2	–	–	–	–	–	–	2	3	3
Average	3	2.4	2.4	2	2.2	0.6	0.6	0.6	0.8	0.6	0.6	2.2	2.8	3

IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name	Applied Algebra	Linear
	Program:	B.Tech. (CSE)	Semester:	I	Course Code		
					L-T-P-C	2-1-0-3	
					Prerequisite Subject	Intermediate Mathematics	
Course Objectives		1. To introduce students to the fundamental concepts of matrices, their properties.					
		2. To develop an understanding of vector spaces, their subspaces, and linear transformations.					
		3. To provide an in-depth understanding of eigenvalues, eigenvectors, and diagonalization, and their geometric interpretations.					
		4. To explore advanced topics in linear algebra, such as Singular Value Decomposition (SVD) and Principal Component Analysis (PCA)					

Course Outcome (At the end of the course, students will be able to)		Description		
CO1		Understand the properties of determinants, rank, and matrix factorizations, and their role in solving linear systems.		
CO2		Identify fundamental concepts of vector spaces, linear transformations, and matrix operations.		
CO3		Apply eigenvalues, eigenvectors, and the Cayley-Hamilton theorem to analyze and factorize matrices in mathematical and engineering contexts.		
CO4		Understand Singular Value decomposition and Principal Component Analysis (PCA).		
Course Description		This course covers essential topics in linear algebra, including matrices, determinants, eigen values, eigenvectors, and vector spaces. Students will explore matrix operations such as Gaussian elimination, Cramer’s rule, and LU decomposition, with a focus on solving systems of linear equations. The course also introduces advanced concepts such as Singular Value Decomposition (SVD) and Principal Component Analysis (PCA) for data analysis.		
Detailed Syllabus				
Unit	Topics		CO No.	Number of lectures
Unit 1: Introduction to Matrices	Introduction to Matrices, Types of matrices, Determinant, Properties of determinant, Elementary row operations and row echelon form, Rank of a matrix and its properties, Inverse of a matrix and its computation.		CO 1	8
Unit 2: System of Linear Equations	Matrix representation of systems of linear equations: consistency and solutions, Cramer rule, Gauss elimination and Gauss-Jordan method, LU decomposition, Operation counts of the methods.		CO 1	8
Unit 3: Vector Space	Introduction of vector space with their basic properties, Subspaces, Linear dependence and Independence of vectors, Linear combinations and span, Basis, Dimension, Linear transformations, Rank and nullity of a transformation, matrix representation of a linear transformation.		CO 2	10
Unit 4: Eigen values and Eigen Vectors	Characteristic equation and polynomial, Eigenvectors and eigenvalues of matrices with geometrical interpretations, Cayley-Hamilton Theorem. Similar matrices and diagonalization.		CO 3	10
Unit 5: SVD and PCA	Inner product space, orthogonal/orthonormal bases, Singular Value Decomposition (SVD), Principal Component Analysis with applications.		CO 4	9
Text Books :				

1. Seymour Lipschutz, Marc Lipson. 2009. "Linear Algebra", Schaum's outlines, ISBN- 978-0-07-154353-8.
2. David C. Lay. 2007. "Linear Algebra and Its Applications", Pearson Education, ISBN- 978-0-17-58333-5.

Reference Books :

1. S.H. Friedberg, A.J. Insel, L.E. Spence. 2004. "Linear Algebra, 4th Edition", PHI Learning.
2. S. Kumaresan. 2001. "Linear Algebra-A Geometric approach", PHI Learning.

Online Resources:

1. Linear Algebra by Prof. Pranav Haridas | Kerala School of Mathematics
<https://archive.nptel.ac.in/courses/111/106/111106135/>
2. Linear Algebra by Prof. Dilip P. Patil | IISc Bangalore
https://onlinecourses.nptel.ac.in/noc22_ma45/preview

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination Examination Scheme:	
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Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)	

[illegible]

IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name	Environmental Studies & Disaster Management
	Program:	B.Tech. (CSE)	Semester:	I	Course Code	CSE26106
					L-T-P-C	2-0-0-2
					Prerequisite Subject	Basics of Environmental science
Course Objectives		1. Understand the environment, ecosystems, natural resources, biodiversity, and their conservation.				
		2. Learn about causes, effects, and control of environmental pollution and the impact of population growth.				
		3. Study types, causes, and impacts of disasters in India and globally, along with risk reduction approach.				
		4. Know disaster management cycles, national policies, and roles of various agencies in disaster mitigation and recovery.				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		To demonstrate understanding of environmental concepts, biodiversity, and conservation practices.				
CO2		To analyze causes and effects of environmental pollution and population growth on ecosystems.				
CO3		To identify different types of disasters, their causes, and impacts at national and global levels.				
CO4		To apply knowledge of disaster management cycles, policies, and stakeholder roles for effective risk reduction and response.				
Course Description		This course provides foundational knowledge of the environment, ecosystems, natural resources, and biodiversity, emphasizing their conservation and sustainable management. It covers the causes, effects, and control of various environmental pollution types and examines the impact of population growth on natural systems. Students will explore the nature, causes, and impacts of disasters both in India and globally, alongside strategies for disaster risk reduction and safety measures. The course also introduces disaster management cycles, relevant national policies, and the roles of key agencies in effective disaster mitigation and post-disaster response.				

Detailed Syllabus

Unit	Topics	CO No.
Unit 1: Environmental Concerns and Sustainable Resource Management	Environmental Concerns, Population Growth, and Resource Consumption, Principles of Sustainable Development, Industrial Ecology & Circular Economy	CO1
Unit 2: Pollution and Its Impact on Environment	Understanding Air Pollution, Climate Change: Scientific Basis, Mitigation & Adaptation, Noise Pollution, Ecology and Soil Pollution	CO2
Unit 3: Renewable Energy, Waste Management, and Environmental Policies	Renewable Energy – I (Solar Energy & Wind Energy), – II (Geothermal Energy, Ocean Energy, Hydroelectricity, Bioenergy), Water Quality, Reuse and Recycling, Solid Waste Management (including E-waste Management), Environmental Policies and Legislation	CO2
Unit 4: Understanding Disaster, Hazards and Vulnerability	Concept of Disaster - Different approaches- Concept of Risk - Levels of Disasters - Disaster Phenomena and Events (Global, national and regional), Natural and man-made hazards; response time, frequency and forewarning levels of different hazards - Characteristics and damage potential or natural hazards; hazard assessment - Dimensions of vulnerability factors; vulnerability assessment - Vulnerability and disaster risk - Vulnerabilities to flood and earthquake hazards,	CO3
Unit 5: Disaster Management Mechanism	Concepts of risk management and crisis managements - Disaster Management Cycle - Response and Recovery - Development, Prevention, Mitigation and Preparedness - Planning for Relief.	CO4

Text Books :

1. Introduction to Environmental Engineering and Science by Gilbert M. Masters and Wendell P. Ela (2020).
2. Environmental Science (16th Edition) by G. Tyler Miller, Jr. & Scott E. Spoolman (2017).
3. Manual on Disaster Management, National Disaster Management, Agency Govt of India. 2. Disaster Management by Mrinalini Pandey Wiley 2014.
4. Disaster Science and Management by T. Bhattacharya, McGraw Hill Education (India) Pvt Ltd Wiley 2015.

Reference Books:

1. Living in the Environment: Concepts, Connections and Solutions (19th Edition) by G. Tyler Miller, Jr. & Scott E. Spoolman (2017)
2. Earth and Atmospheric Disasters Management, N. Pandharinath, CK Rajan, BS Publications 2009.
3. National Disaster Management Plan, Ministry of Home affairs, Government of India (<http://www.ndma.gov.in/images/policyplan/dmplan/draftndmp.pdf>).

Online Resources:

1. Prof. Sudha Goel, "Environmental Science," NPTEL Course, Offered by IIT Kharagpur. [Online]. Available: https://onlinecourses.nptel.ac.in/noc25_hs218/preview.
2. Dr. Naveen Kumar, "Disaster Management," NPTEL Course, Offered by University of Hyderabad. [Online]. Available: https://onlinecourses.swayam2.ac.in/cec19_hs20/preview.

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination

Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	P O 1	P O 2	P O 3	P O 4	P O 5	P O 6	P O 7	P O 8	P O 9	PO 10	P O 11	PO 12	PS O1	PS O2
CO1	2	2	2	–	1	2	3	2	–	2	–	2	1	–
CO2	2	3	2	2	2	2	3	2	–	2	–	2	1	–
CO3	2	2	3	2	2	2	3	2	–	2	2	2	2	1
CO4	2	2	2	1	–	2	3	3	2	3	2	2	–	–
Average	2.0	2.25	2.25	1.75	1.67	2.0	3.0	2.25	2.0	2.25	2.0	2.0	1.33	0.33

IILM University, Greater Noida	School	School of Computer Science & Engineering	Year:	I	Course Name	Professional Communication for Engineers	
					Course Code	CSE26105	
	Program :	B.Tech. (CSE)	Semester :	I	L-T-P-C	2-0-0-2	
					Prerequisite Subject	-	
Course Objectives		1. Understand the fundamental principles of professional communication in engineering contexts.					
		2. Apply verbal, non-verbal, and written communication skills effectively in technical and workplace environments.					
		3. Analyze communication scenarios in professional engineering settings and suggest appropriate strategies.					
		4. Create professional documents and deliver impactful presentations using effective communication techniques.					
Course Outcome (At the end of the course, students will be able to)		Description					
CO1		Demonstrate proficiency in oral, written, and interpersonal communication in technical contexts.					
CO2		Evaluate and improve communication strategies in teams and professional documentation					
CO3		Design persuasive presentations, reports, and proposals using appropriate technical vocabulary and style.					
Course Description		The course <i>Professional Communication for Engineers</i> is designed to develop essential communication skills required in professional and technical environments. It focuses on effective verbal, non-verbal, and written communication, addressing common barriers and strategies to overcome them. Students will learn to craft professional documents, deliver impactful presentations, and collaborate efficiently in team settings. Through practical exercises and real-world scenarios, the course enhances oral, written, and					

	interpersonal communication skills, enabling students to create persuasive reports, proposals, and presentations using appropriate technical language and style for diverse engineering roles.		
Detailed Syllabus			
Unit	Topics	CO No.	Number of lectures
Unit 1: Introduction to Professional Communication	Definition, scope, and importance in engineering. Communication cycle and barriers to communication. Communication in multicultural and global workplaces. Ethics in engineering communication.	CO1	4
Unit 2: Oral Communication	Public speaking and technical presentations, Elevator pitch and briefing skills. Listening and feedback techniques. Telephone and virtual communication etiquette	CO1, CO2	6
Unit 3: Written Technical Communication	Email, memos, business letters, and reports. Writing technical descriptions, instructions and proposals. Editing and proofreading skills, Grammar, clarity and tone.	CO1, CO3	6
Unit 4: Team and Interpersonal Communication	Group dynamics and collaboration. Conflict resolution and negotiation. Leadership and interpersonal skills. Cross-cultural communication	CO2, CO3	4
Unit 5: Digital and Visual Communication	Visual aids: graphs, charts, infographics. Presentation tools: PowerPoint, Prezi, Canva. E-portfolios and LinkedIn profiles. Digital professionalism and netiquette	CO3	4
Unit 6: Capstone Communication Project	Integrative group project simulating an engineering communication scenario (e.g., a client pitch, stakeholder presentation, proposal defense). Final oral and written presentation with Q&A	CO3	6
Text Books:			
1. Meenakshi Raman & Sangeeta Sharma, <i>Technical Communication: Principles and Practice</i> , Oxford University Press.			
2. M. Ashraf Rizvi, <i>Effective Technical Communication</i> , McGraw-Hill.			
3. Lesikar, Raymond V., <i>Business Communication: Making Connections in a Digital World</i> , McGraw-Hill.			
Reference Books :			
1. Anderson, Paul V., <i>Technical Communication: A Reader-Centered Approach</i> , Cengage Learning.			
2. Sharma, S., & Mishra, B. (2015). <i>Communication Skills for Engineers and Scientists</i> . PHI Learning.			
3. Goleman, D. (1998). <i>Working with Emotional Intelligence</i> . Bantam Books.			
4. Lucas, S. E. (2014). <i>The Art of Public Speaking</i> (11th ed.). McGraw-Hill.			
5. Penrose, J. M., & Rasberry, B. (2007). <i>Business Communication for Managers</i> . Thomson South-Western.			
Online Resources:			
1. Purdue Online Writing Lab (OWL): https://owl.purdue.edu/			
2. NPTEL Course: <i>Technical English for Engineers</i> by IIT Madras: https://archive.nptel.ac.in/courses/109/106/109106094/IIT Patna+2NPTEL Online Courses+2NPTEL Online Courses+2NPTEL Online Courses+1NPTEL Online Courses+1GeosNPTEL+2NPTEL+2NPTEL Online Courses+2			

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination Examination Scheme:														
Components		MSE		Presentation/Assignment/ etc						ESE				
Weightage (%)		20%		30%						50%				
Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)														
CO	PO 1	P O 2	P O 3	PO 4	PO 5	P O 6	PO 7	PO 8	PO 9	P O 10	P O 1 1	P O 12	P S O 1	P S O 2
CO1	1	-	2	-	2	-	-	-	-	-	1	-	-	2
CO2	1	2	2	-	2	-	-	-	-	-	1	-	-	2
CO3	1	-	2	-	2	-	-	-	-	-	1	-	-	2
Average	1	0.6	2	-	2	-	-	-	-	-	1	-	-	2

IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name	<i>Foundations of AI & Automation</i>
	Program :	B.Tech. (CSE)	Semester :	I	Course Code	
				L-T-P-C	2-0-0-2	
					Prerequisite Subject	None
Course Objectives		1. To introduce the fundamental concepts, history, and applications of Artificial Intelligence and automation.				
		2. To develop an understanding of basic AI techniques, logic-based reasoning, and automation systems with real-life applications.				
		3. To expose students to foundational tools and real-world applications of AI and automation, encouraging curiosity and innovation in intelligent system design				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		To understand and describe the scope, domains, and evolution of AI and automation technologies.				
CO2		To apply basic logic and reasoning techniques for solving AI-related problems.				
CO3		To understand components, structure, and functioning of basic robotic and automated systems.				
CO4		To analyze current applications and emerging trends in AI and robotics, along with their ethical and societal implications.				

Course Description	This course provides an introductory overview of Artificial Intelligence and automation, covering fundamental concepts, techniques, and real-world applications. Students will learn about intelligent agents, basic problem-solving methods, and emerging AI trends. The course aims to build a strong foundation for further studies in intelligent systems and automation technologies.		
Detailed Syllabus			
Unit	Topics	CO No.	Number of lectures
Unit 1: Introduction Artificial Intelligence	Definition and scope of Artificial Intelligence (AI), Historical context and evolution of AI-Turing Test and Chinese room argument, Problem reduction State space search: Problem-solving algorithm, Major Components of AI and Search Process	CO1	6
Unit 2: Intelligent Agents and Problem Solving	Introduction to Agents and environment, nature of environment and agents, problem solving agent, Search Algorithm: Brute Force Search, Depth First Search, Breadth First Search, Heuristic Search Algorithm: Hill Climbing Algorithm, Best First Search, A and AO* Algorithm, Knowledge Representation	CO2	7
Unit 3: Introduction to Robotics and Automation	Introduction to robotics, Basic Concepts, History and Classification, Types of Robots and Automation Systems like Industrial robots, mobile robots, humanoid robots, Smart appliances, Key Components: Sensors, Actuators, Controllers, Introduction to Control Systems, Automation in Manufacturing, Smart Homes, and Agriculture	CO2, CO3	9
Unit 4: Applications of Robotics and Future Trends	Aerial Robotics: Introduction to Drones (UAVs), Components: Frame, motors, propellers, flight controller, Flight principles: Lift, thrust, drag, Applications: Surveillance, mapping, delivery. Emerging Trends in Robotics: Swarm Robotics, Collaborative Robots (Cobots), Soft Robotics, Human-Robot Interaction (HRI)	CO3, CO4	9
Text Books :			
1. Machine Learning, Tom Mitchell, McGraw Hill, 1997. 2. S. Russell and P. Norvig, “Artificial Intelligence: A Modern Approach”, Prentice Hall, Third Edition, 2009. 3. "Introduction to Robotics: Mechanics and Control" by John J. Craig			
Reference Books :			

1. Bratko, I. (2011). *Prolog: Programming for Artificial Intelligence* (4th ed.). Addison-Wesley Educational Publishers Inc.
2. Janakiraman, V. T., Sarukesi, K., & Gopalakrishnan, P. (2001). *Foundations of Artificial Intelligence and Expert Systems*. New Delhi: Macmillan India Ltd.
3. Introduction to AI Robotics" by Robin R. Murphy
4. Drone Fundamentals: Theory and Practice for Drone Pilots by David G. Price

Online Resources:

1. <https://pll.harvard.edu/course/cs50s-introduction-artificial-intelligence-python>
2. <https://ocw.mit.edu/>

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination

Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

[illegible]

IILM University, Greater Noida	School	School of Computer Science & Engineering	Year:	I	Course Name	<i>Programming in C</i>
	Program:	B.Tech. (CSE)	Semester:	I	Course Code	CSE26108
					L-T-P-C	3-0-0-3
					Prerequisite Subject	Simple Mathematics & basic comprehension skills.
Course Objectives		1. To introduce students to the basic principles of algorithm development using arithmetic, logical operations, and control structures such as loops and decision-making constructs.				
		2. To enable students to understand and implement fundamental data structures like arrays, structures, and unions for solving problems involving data organization.				
		3. To develop students’ ability to design modular programs by decomposing problems into functions.				
		4. To familiarize students with the concept of pointers, files and their use in efficient memory management and dynamic data manipulation.				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		Apply fundamental programming constructs such as loops, conditional statements, arithmetic and logical operations to develop simple algorithms for solving computational problems.				
CO2		Analyze and implement arrays, structures, and unions to solve problems effectively.				
CO3		Design modular solutions by decomposing complex problems into functions and synthesize complete programs.				
CO4		Apply the concept of pointers for efficient memory management and dynamic data handling and working with files.				
Course Description		This course introduces the fundamental concepts of programming using a high-level language. It covers algorithm development, basic input/output operations, control structures, and data representation using arrays, structures, and unions. Emphasis is placed on problem-solving through logical and arithmetic				

	operations, function-based modular programming, and the application of pointers for memory management. Students will gain hands-on experience in writing, testing, and debugging programs while learning to apply programming principles to solve real-world problems efficiently. The course also encourages structured problem decomposition and algorithmic thinking to solve real world problems.			
Detailed Syllabus				
Unit	Topics	CO No.	Number of lectures	
Unit 1: Computer Fundamentals	Introduction of Computers, Classification of Computers, Anatomy of a Computer, Memory Hierarchy, Introduction to OS, Operational Overview of a CPU. Program Fundamentals: Generation and Classification of Programming Languages, Compiling, Interpreting, Loading, Linking of a Program, Developing Program, Software Development. Algorithms: Definitions, Different Ways of Stating Algorithms (Step-form, Pseudo-code, Flowchart), Strategy for Designing Algorithms, Structured Programming Concept. Basics of C: Overview of C, Developing Programs in C, Parts of Simple C Program, Structure of a C Program, Comments, Program Statements, C Tokens, Keywords, Identifiers, Data Types, Variables, Constants, Operators and Expressions, Expression Evaluation–precedence and associativity, Type Conversions.	CO 1	11	
Unit 2: Input-Output	Non-formatted and Formatted Input and Output Functions, Escape Sequences, Control Statements: Selection Statements – if, if-else, nested if, nested if-else, comma operator, conditional operator, switch; Iterative Statements–while, for, do-while; Special Control Statement–goto, break, continue, return, exit.	CO 2	9	
Unit 3: Arrays	Array Notation and Representation, Manipulating Array Elements, using Multi-Dimensional Arrays. Character Arrays and Strings, Dynamic Memory Allocation. Introduction to file handling.	CO 3	5	
Unit 4: Functions	Introduction, Types of Functions, Functions with Array, Passing Parameters to Functions, Call by Value, Call by Reference, Passing Arrays to Functions. Recursive Functions.	CO 4	5	
Unit 5: Pointers	Declarations – Passing pointers to functions – Operation in Pointers - Pointers Arrays – Arrays of pointers – Structures and Union	CO 4	5	
Unit 6: User-defined Data Types	Declaring a Structure (Union) and its members, Initialization Structure (Union), Accessing members of a Structure (Union), Array of Structures (Union), Structures verses Unions, Enumeration Types. Files: Introduction, Using Files in C, Working with Text Files, Working with Binary Files, Files of Records, Random Access to Files of Records, Other File Management Functions.	CO 3, CO 4	10	
Text Books :				

1. B.W. Kernighan and D.M. Ritchie, The C Programming Language, 2nd Edition, PHI, 1988.
2. Kanethkar Y., Let us C, BPB Pub., New Delhi. 1999.

Reference Books :

1. Computer Concepts and Programming in C, E Balaguruswami, McGraw Hill
2. Computer Concepts and Programming in C, R.S. Salaria, Khanna Publishing House

Online Resources:

1. NPTEL – Programming in C (Prof. Anupam Basu, IIT Kharagpur)

<https://nptel.ac.in/courses/106105085>

2. GeeksforGeeks – C Programming Language

<https://www.geeksforgeeks.org/c-programming-language/>

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination

Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	3	2	2	-	2	-	-	-	-	-	-	1	3	-
CO2	3	2	2	1	3	-	-	-	-	-	-	1	3	-
CO3	3	2	2	1	3	-	-	-	-	-	-	1	3	-
CO4	3	2	2	1	3	-	-	-	1	-	-	1	3	-
Average	3	2	2	0.75	2.75	-	-	-	0.25	-	-	1	3	-

 IILM University, Greater Noida	School	School of Computer Science & Engineering	Year:	I	Course Name	Applied Calculus
	Program :	B.Tech. (CSE)	Semester:	II	Course Code	CSE26103
					L-T-P-C	2-1-0-3
					Prerequisite Subject	Intermediate Mathematics
Course Objectives		1. To introduce the fundamental concepts of limits, continuity, differentiability, and expansion theorems for functions of one and two variables. 2. To develop skills in evaluating partial derivatives, total derivatives, and solving optimization problems.				

	3. To familiarize students with techniques of single, double, and triple integration and their applications in computing areas and volumes.		
	4. To apply calculus concepts in optimization methods such as gradient descent, Newton's method, and multiple linear regression.		
Course Outcome (At the end of the course, students will be able to)	Description		
CO1	Define fundamental concepts of limits, continuity, differentiability, and standard theorems in single and multivariable calculus.		
CO2	Understand the concepts of total derivative and the use of derivative for language multiplier method.		
CO3	Apply integration techniques to solve real-world problems involving multiple integrals.		
CO4	Examine the role of calculus in optimization problems by evaluating gradient vectors, Hessian matrices, and numerical methods like gradient descent.		
Course Description	This course provides a comprehensive foundation in differential and integral calculus for single and multiple variables. Topics include limits, continuity, differentiation, Taylor and Maclaurin series expansions, partial derivatives, and multivariable optimization. Students will also study definite and indefinite integrals, double and triple integrals, and their applications to real-world problems.		
Detailed Syllabus			
Unit	Topics	CO No.	Number of lectures
Unit 1: Single variable Calculus	Introduction to Limits, Continuity, and Differentiability for one variable, Indeterminate forms, Rolle's theorem, Lagrange's mean value theorem, Successive differentiation, Taylor's & Maclaurin's expansion for the functions of one variable.	CO1	9
Unit 2: Two Variable Calculus	Introduction to Limits and Continuity for 2 variables, Partial derivatives and gradient vector, Differentiability of functions of several variables, Taylor & Maclaurin Series for 2 variables.	CO1	9
Unit 3: Minima and Maxima in 2 variables	Total derivatives, homogeneous functions, Euler's theorem, Maxima and minima for the function of two variables, Lagrange's method of multipliers for constrained extrema.	CO2	9
Unit 4: Integral calculus	Indefinite integrals and standard formulas, Definite integrals and properties, Fundamental Theorem of Calculus, Double integral, Triple integral, change of order of integration and change of variables, Application to calculation of area and volumes.	CO3	9
Unit 5: Applications to Optimization problems	Role of calculus in optimization problems, Gradient vector and Hessian matrices, Gradient descent algorithm, Newton's method for optimization, Simple and Multiple Linear regression through optimization.	CO4	9
Text Books :			
1. 1. Khalil Ahmad and Pankaj Sharma. 2022. "Text Book of Calculus". New Age International Publishers, ISBN- 978-9393159991			
2. Hughes-Hallett et al..2001. "Calculus- Single and Multivariable (3rd Edition)". John-Wiley & Sons, ISBN- 978-0471408277			
Reference Books :			

1. James Stewart. 2009. "Single variable Calculus: Concepts and Contexts", 4th Edition. Cengage Learning, ISBN-978-0495-55972-6.
2. Peter D. Lax and Maria S. Terrell. 2017. "Multivariable Calculus with Applications", Springer, ISBN 978-3319-74072-0.
3. Jorge Nocedal, Stephen J. Wright. 1999. "Numerical Optimization". Springer, ISBN: 978-0387-30303-1.

Online Resources:

1. Basic Calculus – 1 By Prof. Arindama Singh | IIT Madras

https://onlinecourses.nptel.ac.in/noc21_ma16/preview

2. Multivariable Calculus. By Prof. S. K. Gupta, Prof. Sanjeev Kumar | IIT Roorkee

https://onlinecourses.nptel.ac.in/noc24_ma33/preview

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination

Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	2	1	-	-	1	-	-	-	-	-	-	1	-	-
CO2	2	1	-	-	-	-	-	-	-	-	-	1	-	-
CO3	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO4	3	1	-	-	2	-	-	-	-	-	-	-	-	-
Average	2.5	1	-	-	1	-	-	-	-	-	-	0.5	-	-

	School:	School of Computer Science & Engineering	Year:	I	Course Name:	Applied Calculus Lab
					Course Code:	CSE26103P
	Program:	B. Tech. (CSE)	Semester:	II	L-T-P-C	0-0-2-1
					Prerequisite Subject:	
Course Objectives			1. To develop computational skills for exploring foundational concepts in limits, continuity, derivatives, and series expansions.			

	2. To apply multivariable calculus concepts such as gradients, optimization, and constrained optimization through programming. 3. To build the ability to approximate functions using Taylor and Maclaurin series and analyze the behavior of functions graphically. 4. To introduce students to practical optimization techniques such as Gradient Descent and Newton's Method using real-world datasets.
Course Outcome (At the end of the course, students will be able to)	Description
CO1	Understand key calculus concepts such as limits, continuity, derivatives, and Taylor series numerically and graphically.
CO2	Apply methods of multivariable calculus, including partial derivatives for optimization.
CO3	Implement and analyze optimization algorithms such as Gradient Descent and Newton's Method for solving linear regression problems.
Course Description	This lab introduces students to key concepts of calculus through programming-based exploration. Students will investigate limits, continuity, derivatives, Taylor and Maclaurin series, and optimization methods in one and several variables. Practical exercises focus on using Python to visualize functions, perform symbolic computations, and solve real-world problems involving gradients, maxima-minima, constrained optimization, and regression.
List Of Experiments	1.Numerical exploration of limit and continuity To understand the concept of limits and continuity by evaluating a function as the input approaches a specific value and visualizing its behaviour graphically. 2. Graphing functions and derivatives To visualize the relationship between a function and its derivative. 3. Taylor and Maclaurin Series – Symbolic Computation To compute Taylor and Maclaurin series of elementary functions up to a specified degree and analyze the error. 4. Gradient Vectors and Their Geometrical Meaning To compute and visualize gradient vectors of functions of two variables. 5. Homogenous functions, Euler's theorem and Multivariable Taylor series To identify homogeneous functions, verify Euler's Theorem and compute Taylor series for multivariable functions. 6. Minima and Maxima for one and two variable functions To locate critical points of functions of one and two variables using first and second derivatives tests. 7. Lagrange Multipliers for Constrained Optimization To solve constrained optimization problems using the method of Lagrange multipliers. 8. Gradient Descent for Function Minimization To implement the gradient descent algorithm to find minima of multivariable functions.

	9. Newton's method for Function Minimization To implement the Newton's method to find minima of multivariable functions. 10. Multiple Linear Regression using Gradient Descent To implement multiple linear regression using optimization via gradient descent.
Experiment 1:	Title: Numerical exploration of limit and continuity Objective: To understand the concept of limits and continuity by evaluating a function as the input approaches a specific value and visualizing its behavior graphically. Lab Activities: • Numerical evaluation of limits: Write a program to numerically evaluate one-sided and two-sided limits. • Graphical visualization: Plot the function and highlight the limiting behavior near the chosen point and explore removable, jump and infinite discontinuities.
Experiment 2:	Title: Graphing functions and derivatives Objective: To visualize the relationship between a function and its derivative. Lab Activities: • Plotting a function and its derivative: Write a program to plot a function and its first derivative on the same graph. • Deriving insights from the plot: Analyze intervals of increase/decrease, maxima, minima, and inflection points.
Experiment 3:	Title: Taylor and Maclaurin Series – Symbolic Computation Objective: To compute Taylor and Maclaurin series of elementary functions up to a specified degree and analyze the error. Lab Activities: • Computing Taylor and Maclaurin expansions: Write a program to compute Taylor or Maclaurin expansion of a function using formula or built-in tools. • Analyzing function value and expansions: Compare function values with truncated series at various points and analyze approximation error with respect to the degree of the computed polynomial.
Experiment 4:	Title: Gradient Vectors and Their Geometrical Meaning Objective: To compute and visualize gradient vectors of functions of two variables. Lab Activities: • Computing gradients at selected points Write a program to compute gradient vector of a two-variable function at certain selected points.

	● Visualization of gradient: Plot the surface and overlay gradient vectors as arrows and thus relate gradient direction to steepest ascent.
Experiment 5:	Title: Homogenous functions, Euler's theorem and Multivariable Taylor series Objective: To identify homogeneous functions, verify Euler's Theorem and compute Taylor series for multivariable functions. Lab Activities: ● Homogenous functions and Euler's theorem: Write a program to compute partial derivatives of a function and test it for homogeneity and verify the Euler's theorem. ● Multivariable Taylor series: Use Taylor series to expand multi-variable functions and explore the error and accuracy.
Experiment 6:	Title: Minima and Maxima for one and two variable functions Objective: To locate critical points of functions of one and two variables using first and second derivatives tests. Lab Activities: ● Minima/Maxima for one variable functions: Write a program to compute local minima/maxima for one variable functions. ● Minima/Maxima for two variable functions: Write a program to compute local minima/maxima for two variable functions and visualize surface and critical points using 3D plots.
Experiment 7:	Title: Lagrange Multipliers for Constrained Optimization Objective: To solve constrained optimization problems using the method of Lagrange multipliers. Lab Activities: ● Deploying the method of Lagrange Multipliers for constrained optimization Write a program to compute extrema of a function $f(x,y)$ subject to $g(x,y)=c$. ● Visualize constraint curve and critical points Plot the constraint curve $g(x,y)=c$ and visualize the obtained critical points.
Experiment 8:	Title: Gradient Descent for Function Minimization Objective: To implement the gradient descent algorithm to find minima of multivariable functions. Lab Activities: ● Implementing gradient descent Write a program that defines a cost function, computes gradient and then updates the parameter values to arrive at the minimum value of the function. ● Understand the role of learning rate

	Use different learning rate to understand how quickly or slowly the algorithm converges to the minimum value of the underlying function.						
Experiment 9:	Title: Newton’s method for Function Minimization Objective: To implement the Newton’s method to find minima of multivariable functions. Lab Activities: ● Implementing Newton’s method Write a program that defines a cost function, computes gradient, and hessian, and then updates the parameter values to arrive at the minimum value of the function. ● Understand the role of learning rate Use different learning rate to understand how quickly or slowly the algorithm converges to the minimum value of the underlying function.						
Experiment 10:	Title: Multiple Linear Regression using Gradient Descent Objective: To implement multiple linear regression using optimization via gradient descent. Lab Activities: ● Implementing Multiple Linear Regression Write a program to prepare a small dataset with multiple attributes, define loss function and update the model parameters using Gradient descent algorithm. ● Analyzing the results: Compare the model parameters learned using the gradient descent with built-in regression tools.						
Text Books:							
1. Khalil Ahmad and Pankaj Sharma. 2022. “Text Book of Calculus”. New Age International Publishers, ISBN- 978-9393159991 2. Hughes-Hallett et al.. 2001. “Calculus- Single and Multivariable (3rd Edition)”. John-Wiley & Sons, ISBN- 978-0471408277							
Reference Books :							
1. James Stewart. 2009. “Single variable Calculus: Concepts and Contexts”, 4th Edition. Cengage Learning, ISBN- 978-0495-55972-6. 2. Peter D. Lax and Maria S. Terrell. 2017. “Multivariable Calculus with Applications”, Springer, ISBN 978-3319-74072-0. 3. Jorge Nocedal, Stephen J. Wright. 1999. “Numerical Optimization”. Springer,ISBN: 978-0387-30303-1.							
Online Resources:							
1. Basic Calculus – 1 By Prof. Arindama Singh IIT Madras https://onlinecourses.nptel.ac.in/noc21_ma16/preview 2. Multivariable Calculus. By Prof. S. K. Gupta, Prof. Sanjeev Kumar IIT Roorkee https://onlinecourses.nptel.ac.in/noc24_ma33/preview							
Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination							
Examination Scheme:							
<table><tr><td>Components</td><td>Continuous Assessment</td><td>Lab Test / Presentation / Assignment / etc.</td></tr><tr><td>Weightage (%)</td><td>50%</td><td>50%</td></tr></table>		Components	Continuous Assessment	Lab Test / Presentation / Assignment / etc.	Weightage (%)	50%	50%
Components	Continuous Assessment	Lab Test / Presentation / Assignment / etc.					
Weightage (%)	50%	50%					

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)														
CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	2	1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO3	2	1	-	-	2	-	-	-	-	-	-	-	-	-
Average	2.3	1	-	-	1	-	-	-	-	-	-	-	-	-

 IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name:	<i>Programming in C Lab</i>
					Course Code:	CSE26108P
	Program:	B. Tech. (CSE)	Semester:	I	L-T-P-C	0-0-2-1
Course Objectives						
		1. To develop proficiency in using fundamental C programming constructs such as variables, data types, control structures, arrays, functions, pointers, structures, and file handling.				
		2. To enhance students' problem-solving and logical reasoning skills through hands-on implementation of modular and efficient C programs.				
		3. To enable students to design, write, test, and debug C programs for solving real-world problems and building foundational software development skills.				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		Apply the branching and looping constructs to execute the basic C programs.				
CO2		Execute the programs based on Array and Structure,				
CO3		Execute programs based on functions and pointers.				
Course Description		The Programming in C Lab is designed to complement the theoretical knowledge of C programming with practical implementation skills. This lab focuses on developing students' proficiency in writing, compiling, and debugging C programs using a				

Commented [1]:

	structured approach. Students will work on a variety of programming tasks involving decision-making, loops, arrays, strings, functions, pointers, structures, and file handling. The course encourages logical thinking and systematic problem-solving, preparing students to apply programming concepts effectively to real-world scenarios and laying a strong foundation for advanced computing courses.
List Of Experiments	1. Basic Data Types To demonstrate the use of various data types and display their memory size and value ranges.
	2. Basic Arithmetic Operations To compute the sum and average of three numbers using user input.
	3. Swapping of Variables To swap the values of two variables using both third-variable and arithmetic operations.
	4. Area of Circle To calculate the area of a circle from radius input and display the result in different notations.
	5. Conditional Statements To apply decision-making using if-else constructs for leap year checking, comparisons, and finding even/odd or greatest numbers.
	6. Grading System To read marks, compute percentage, and assign grades based on predefined criteria.
	7. Switch Case and Summation To perform arithmetic operations using switch-case and compute the sum up to a given number.
	8. Loops and Recursion To practice loops and recursion for factorial, Fibonacci series, and even/odd sum calculations.
	9. Functions and Number Theory To define and use functions for checking prime, finding reverse, calculating factorial, and printing Armstrong numbers.
	10. Arrays and Functions To perform operations like sum, search, and comparison on arrays using user-defined functions.
	11. Sorting and Matrices To sort an array using Bubble Sort and perform matrix addition, multiplication, and diagonal summation.
	12. Strings, Structures, and Pointers

	To manipulate strings, use structures for train scheduling, and swap variables using pointers.
Experiment 1:	Title: Basic Data Types and Their Ranges Objective: To print various data types and display their memory size and value ranges. Lab Activities: Write a C program to print different data types and their ranges using <code>sizeof()</code> and <code>limits.h/float.h</code>.
Experiment 2:	Title: Basic Arithmetic Operations Objective: To compute the sum and average of three numbers. Lab Activities: Write a program to read three integers and calculate the sum and average
Experiment 3:	Title: Swapping of Variables Objective: To swap values with and without using a third variable. Lab Activities: - Write a C program to swap two numbers using a third variable. - Write a C program to swap two numbers without using a third variable
Experiment 4:	Title: Area of circle Objective: To calculate the area of a circle using radius input and print it in different notations. Lab Activities: Write a program to calculate area using <code>float</code> and display it in normal and exponential format.
Experiment 5:	Title: Conditional Statements (Leap Year, Comparison) Objective: To use if-else and logical operators for condition checking. Lab Activities: - Check whether a year is a leap year. - Check equality of two user-input numbers. - Find the greatest of three numbers. - Determine if a number is even or odd.
Experiment 6:	Title: Grading System Objective: To calculate percentage and assign grades based on marks. Lab Activities: - Read marks for 5 subjects. - Compute percentage. - Display grade using conditional logic.

Experiment 7:	Title: Switch Case and Summation Objective: To implement control structures like switch, loops. Lab Activities: • Perform arithmetic operations based on user input using switch. • Print sum of numbers up to a given number.
Experiment 8:	Title: Loops and Recursion Objective: To apply looping and recursive techniques to solve mathematical problems. Lab Activities: 1. Calculate factorial using loop. 2. Print sum of even and odd numbers. 3. Generate Fibonacci series.
Experiment 9:	Title: Functions and Number Theory Objective: To create user-defined functions to perform number-based computations. Lab Activities: • Check if number is prime. • Find reverse of a number. • Print Armstrong numbers from 1 to 100. • Recursive factorial.
Experiment 10:	Title: Array and Functions Objective: To demonstrate use of arrays and functions for computation. Lab Activities: • Take array input and compute sum. • Add two arrays and store result in third array. • Find min and max in an array. • Perform linear search.
Experiment 11:	Title: Sorting and Matrices Objective: To sort arrays and perform matrix operations. Lab Activities: • Sort array using Bubble Sort. • Add and multiply two matrices. • Find sum of diagonal elements in a matrix
Experiment 12:	Title: Strings, Structures, and Pointers Objective: To implement operations using strings, structures and pointers. Lab Activities: • Implement strlen, strcat, and strcpy using functions. • Define TRAIN_INFO structure and perform operations (filtering by station/time). • Swap two variables using pointers.

Text Books:																																								
1. B.W. Kernighan and D.M. Ritchie, The C Programming Language, 2nd Edition, PHI, 1988.																																								
2. Kanethkar Y., Let us C, BPB Pub., New Delhi. 1999.																																								
Reference Books :																																								
1. Computer Concepts and Programming in C, E Balaguruswami, McGraw Hill																																								
2. Computer Concepts and Programming in C, R.S. Salaria, Khanna Publishing House																																								
Online Resources:																																								
1. GeeksforGeeks – C Programming Language https://www.geeksforgeeks.org/c-programming-language/ A comprehensive guide with tutorials, examples, quizzes, and coding problems ideal for beginners and advanced learners.																																								
2. NPTEL – Programming in C (by Prof. Anupam Basu, IIT Kharagpur) https://nptel.ac.in/courses/106105085 A government-supported video lecture series ideal for academic reinforcement.																																								
3. Javatpoint – C Programming Language Tutorial https://www.javatpoint.com/c-programming-language-tutorial A good reference for syntax, programs, interview questions, and conceptual clarity.																																								
4. CS50’s Introduction to Computer Science (Harvard, C-Based) https://cs50.harvard.edu/x/																																								
Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination																																								
Examination Scheme:																																								
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CO	PO 1	P O2	P O3	PO 4	PO 5	P O6	PO 7	PO 8	PO 9	P O 10	P O 11	P O 12	P S O 1	P S O 2																										
CO1	2	2	1	2	-	-	-	-	-	-	1	3	-	-																										
CO2	2	2	2	2	-	-	-	-	-	-	1	3	-	-																										
CO3	2	2	2	3	-	-	-	-	-	1	-	2	3	-																										
Average	2	2	1.6	2.3	-	-	-	-	-	0.3	0.6	2.6	1	-																										

 IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name:	<i>Linux Administration Lab</i>
					Course Code:	CSE26111
	Program:	B. Tech. (CSE)	Semester:	I	L-T-P-C	0-0-2-1
					Prerequisite Subject:	Programming Basics
Course Objectives		1. To provide practical knowledge of Linux command-line utilities and shell scripting.				
		2. To enable students to manage users, permissions, file systems, and software packages.				
		3. To familiarize students with service and process management in Linux.				
		4. To prepare students for industry-relevant tasks in system and network administration.				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		Use essential Linux commands to navigate, manipulate, and manage the file system.				
CO2		Perform user administration, permission management, and group control operations.				
CO3		Control processes, automate tasks using shell scripts, and manage services.				

CO4	Configure basic networking, SSH access, and understand system logs and package management.
Course Description	This lab-based course introduces students to the fundamentals of Linux system administration. Students will gain hands-on experience with Linux operating system commands, shell scripting, user and permission management, software installation, system services, and network configuration. The lab is designed to build foundational skills required for DevOps, cloud administration, cybersecurity, and system engineering roles.
List Of Experiments	1. Linux Installation and Setup Install Linux using VirtualBox or configure dual-boot with Windows, and perform initial system setup and package updates. 2. System Login and Performance Monitoring Log in via terminal and GUI, and monitor system performance using top, htop, uptime, and vmstat. 3. Exploring Basic Linux Commands Navigate the file system and perform basic file operations using commands like cd, ls, pwd, touch, cp, mv, rm, cat, and more. 4. Using Command Documentation Use man pages and --help options to understand syntax and usage of Linux commands. 5. User and Group Management Add and delete users and groups, assign roles, and switch users using useradd, userdel, groupadd, gpasswd, and su. 6. Managing File Permissions and Ownership Modify file permissions and ownership using chmod, chown, chgrp, and explore default permissions with umask, getfacl, and setfacl. 7. Redirection and Piping in Linux Redirect input/output using >, >>, <, 2>, and combine commands using pipes (). 8. Command Chaining and Logical Operators Chain multiple commands using &&, , and ; to control execution flow. 9. Archiving and Compression Techniques Archive and compress files using tools like tar, gzip, bzip2, xz, and star. 10. Process and Job Control

	Manage active processes using ps, kill, nice, renice, and control background jobs with &, jobs, fg, and bg. 11. Scheduling Tasks in Linux Automate task scheduling using cron and at for periodic and one-time jobs. 12. Introduction to Shell Scripting Write and execute basic shell scripts using variables, user input, output display, and simple conditionals (if, if-else, case). 13. Implementing Loops in Shell Scripts Use for, while, and until loops in shell scripts for iterative task automation. 14. Automated Backup Script Creation Write shell scripts for automated backups using rsync, scp, or tar, and schedule them using cron. 15. Disk, File System, and Network Utilities Manage disks and filesystems with df, du, mount, umount, fdisk, and mkfs, and perform basic network configuration and testing using ip, ping, nmcli, and dig.
Experiment 1:	Title: Linux Installation and Setup Objective: To install and configure Linux using virtualization or dual-boot setup and perform initial system setup. Lab Activities: • Install Linux using VirtualBox or as a dual-boot system. • Perform initial system setup including user creation. • Update and upgrade system packages.
Experiment 2:	Title: Linux Installation and Setup Objective: To install and configure Linux using virtualization or dual-boot setup and perform initial system setup. Lab Activities: • Install Linux using VirtualBox or as a dual-boot system. • Perform initial system setup including user creation. • Update and upgrade system packages.
Experiment 3:	Title: Basic Linux Commands Objective: To explore Linux file system and perform file operations using basic commands. Lab Activities: • Navigate directories using cd, ls, pwd.

	• Create, copy, move, and delete files using touch, cp, mv, rm. • View file contents using cat, more. • Access help using man and --help.
Experiment 4:	Title: User and Group Management Objective: To create, delete, and manage users and groups in a Linux system. Lab Activities: • Add and delete users using useradd, userdel. • Create and manage groups with groupadd, gpasswd. • Assign users to groups and switch users with su.
Experiment 5:	Title: File Permissions and Ownership Objective: To manage file permissions and ownership in Linux. Lab Activities: • Set file permissions using chmod. • Change file ownership using chown and chgrp. • Set default permissions using umask. • Manage ACLs using getfacl and setfacl.
Experiment 6:	Title: Redirection and Piping Objective: To perform input/output redirection and combine commands using pipes. Lab Activities: • Redirect input/output using >, >>, <, 2>. • Combine commands using pipes . • Use command chaining with &&, , ;.
Experiment 7:	Title: Archiving and Compression Objective: To compress and archive files using Linux utilities. Lab Activities: • Archive files using tar, star. • Compress and decompress using gzip, bzip2, xz.
Experiment 8:	Title: Process and Job Control Objective: To manage system processes and background jobs. Lab Activities: 16. List and manage processes using ps, top, kill. 17. Change process priority with nice, renice. 18. Run jobs in background using &, jobs, fg, bg. 19. Schedule tasks using cron and at.
Experiment 9:	Title: Shell Scripting Basics Objective: To write basic shell scripts with variables, inputs, and conditionals.

	Lab Activities: • Create and run a shell script. • Use variables and read user input. • Print outputs using echo. • Implement if, if-else, and case statements.
Experiment 10:	Title: Shell Scripting – Loops Objective: To automate repetitive tasks using loops in shell scripting. Lab Activities: • Write for, while, and until loops. • Combine loops with conditional statements.
Experiment 11:	Title: Backup Automation Objective: To automate backup processes using shell scripts and scheduling tools. Lab Activities: • Write a backup script using rsync, tar, or scp. • Schedule periodic backups using cron.
Experiment 12:	Title: Disk and File System Management Objective: To manage disk space, partitions, and file systems. Lab Activities: • Check disk usage with df and du. • Mount and unmount devices using mount, umount. • Explore partitioning using fdisk and formatting with mkfs. • Briefly explore LVM setup.
Experiment 13:	Title: Service Management. Objective: To manage system services and view logs. Lab Activities: • Start, stop, enable, and disable services using systemctl and service. • View service logs using journalctl. • Rotate logs using logrotate.
Experiment 14:	Title: Network Configuration Objective: To configure and troubleshoot network settings. Lab Activities: • Configure interfaces using ifconfig, ip, and nmcli. • Test connectivity using ping, netstat, ss. • Troubleshoot DNS using nslookup and dig.

Experiment 15:	Title: SSH and Remote Access Objective: To set up secure remote access using SSH.																			
	Lab Activities: ● Install and configure SSH server and client. ● Generate SSH key pairs. ● Set up key-based authentication for secure remote login.																			
	Text Books: 1. Wale Soyinka. Linux Administration: A Beginner's Guide. McGraw-Hill Education. 2. Yashavant Kanetkar. Unix Shell Programming. BPB Publications.																			
	Reference Books : 1. Evi Nemeth, Garth Snyder, Trent R. Hein, Ben Whaley, Dan Mackin. UNIX and Linux System Administration Handbook. Addison-Wesley Professional. 2. Richard Blum. 2021.Linux Command Line and Shell Scripting Bible, 4th Edition, Wiley.																			
Online Resources: 1. Linux Journey – Interactive learning path: https://linuxjourney.com/ 2. The Linux Command Line – Beginner-friendly command guide: https://linuxcommand.org/ 3. Explainshell – CLI command explanation tool: https://explainshell.com/																				
Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination Examination Scheme: <table><tr><td>Components</td><td>Continuous Assessment</td><td>Lab Test / Presentation / Assignment / etc.</td></tr><tr><td>Weightage (%)</td><td>50%</td><td>50%</td></tr></table>															Components	Continuous Assessment	Lab Test / Presentation / Assignment / etc.	Weightage (%)	50%	50%
Components	Continuous Assessment	Lab Test / Presentation / Assignment / etc.																		
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Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)																				
CO	PO 1	P O2	P O3	PO 4	PO 5	P O6	PO 7	PO 8	PO 9	P O 10	P O 11	P O 12	P S O 1	P S O 2						
CO1	1	1	-	-	2	-	-	-	-	-	-	-	3	2						
CO2	1	1	-	-	2	-	-	-	1	-	-	-	3	2						
CO3	1	1	-	-	2	-	-	-	1	-	-	-	3	2						
CO4	1	1	-	-	2	-	-	-	1	-	-	-	3	2						
Average	1	1	-	-	2	-	-	-	0.7	-	-	-	3	2						

IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name	<i>Computational Design Thinking</i>
	Program:	B.Tech. (CSE)	Semester:	I	Course Code	CSE26104
					L-T-P-C	2-1-0-3
					Prerequisite Subject	Basic communication and mathematical reasoning skills
Course Objectives	1. To build a foundational understanding of structured and analytical thinking.					
	2. To enable students to apply problem-solving strategies to real-world and abstract problems.					
	3. To develop logical reasoning, decision-making, and algorithmic thinking.					
	4. To enhance critical and creative thinking through exercises, case studies, and collaborative activities.					
Course Outcome (At the end of the course, students will be able to)	Description					
CO1	Apply structured problem-solving models such as Polya’s method and the IDEAL framework to analyze and solve both well-structured and ill-structured problems.					
CO2	Analyze logical structures, reasoning patterns, and common fallacies to enhance decision-making and argument evaluation.					
CO3	Develop basic algorithms, pseudocode, and flowcharts to model everyday processes using principles of computational thinking.					
CO4	Design creative, ethical, and collaborative solutions to real-world problems using critical thinking, design thinking, and team-based strategies.					
Course Description	This course develops the core cognitive and analytical skills required to approach problems systematically. It blends classical problem-solving strategies, critical thinking tools, and algorithmic reasoning to empower students to address challenges across academic and real-world contexts. Tutorials will focus on hands-on logic exercises, team challenges, and creative thinking workshops.					

Detailed Syllabus

Unit	Topics	CO No.	Number of lectures
Unit 1: Foundations of Computational Design Thinking and Problem Solving	Introduction to Computational Design Thinking, Nature of Problems: Well-Structured vs. Ill-Structured Problems, Characteristics of an Effective Problem Solver, Problem-Solving Models: IDEAL Model and Polya's Four-Step Method, Computational Problem-Solving Process: Understand→ Plan→ Solve→ Verify→ Reflect, Human-Centered Design Principles, Empathy Mapping and User Interviews, Problem Definition: User-Centered Problem Statements, Identifying Functional and Technical Requirements	CO1	9
Unit 2: Creative Thinking and Design Ideation	Creative Thinking in Computational Design, Divergent and Convergent Thinking, Brainstorming and Brainwriting Techniques, SCAMPER Technique, Mind Mapping for Idea Generation, Lateral Thinking Strategies, Design Thinking Process: Empathize, Define, Ideate, Selecting and Prioritizing Ideas	CO1, CO2	9
Unit 3: Computational Problem-Solving using Algorithms	Algorithmic Thinking, Characteristics of Good Algorithms, Pseudocode Writing, Flowcharts, Problem Decomposition, Pattern Recognition, Conditional Logic (If–Else), Iterative Processes (Loops), Basic Data Structures for Problem Solving, Introduction to Algorithm Efficiency and Time Complexity	CO3	9
Unit 4: Critical Thinking and Computational Decision Making	Critical Thinking Fundamentals, Components and Stages of Critical Thinking, Deductive and Inductive Reasoning, Syllogisms, Digital Logic Fundamentals Analogies and Logical Relationships, Cause-and-Effect Analysis, Logical Fallacies and Cognitive Biases, Evaluating Data, Evidence, and Arguments, Decision Making under Uncertainty	CO1, CO2	9
Unit 5: Applications of Computational Design Thinking	Computational Thinking in Real-World Systems, Applying: Abstraction, Decomposition, Pattern Recognition, Algorithmic Thinking, Rapid Prototyping and Iterative Testing, Ethical and Sustainable Computational Design, Design Thinking for Innovation and Entrepreneurship, Computational Design Thinking aligned with Sustainable Development Goals (SDGs)	CO4	9

Text Books:

1. Lau, J. Y. F. (2011). An Introduction to Critical Thinking and Creativity: Think More, Think Better. Wiley.
2. Dowden, B. (2017). Logical Reasoning: A Beginner's Guide. CreateSpace.
- Wing, J. M. (2006). Computational Thinking. Communications of the ACM, 49(3), 33–35.

3. Reference Books :

1. Facione, P. A. (2011). Think Critically. Pearson.
2. De Bono, E. (2015). Lateral Thinking: Creativity Step by Step. Penguin Books.

Dromey, R. G. (2008). How to Solve It by Computer. PHI.
3. Online Resources:
1. MIT OpenCourseWare - Introduction to Problem Solving. https://ocw.mit.edu/courses/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016/ 2. Critical Thinking by the University of Edinburgh – Coursera. https://www.coursera.org/learn/critical-thinking-skills
Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination
Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)														
CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	1	2	1	-	-	-	-	-	-	-	-	1	1	-
CO2	1	2	1	1	-	-	-	-	-	-	-	1	1	-
CO3	2	1	2	-	1	-	-	-	-	-	-	1	1	-
CO4	1	1	2	1	1	-	-	-	1	1	1	1	-	-
Average	1.25	1.5	1.5	.5	.5	-	-	-	.25	.25	.25	1	.75	-

IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name	Foundation of AI & Automation
	Program:	B.Tech. (CSE)	Semester:	I	Course Code	CSE26107
					L-T-P-C	2-0-0-2
					Prerequisite Subject	None
Course Objectives		1. To introduce the fundamental concepts, history, and applications of Artificial Intelligence and automation.				
		2. To develop an understanding of basic AI techniques, logic-based reasoning, and automation systems with real-life applications.				
		3. To expose students to foundational tools and real-world applications of AI and automation, encouraging curiosity and innovation in intelligent system design				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		To understand and describe the scope, domains, and evolution of AI and automation technologies.				
CO2		To apply basic logic and reasoning techniques for solving AI-related problems.				
CO3		To understand components, structure, and functioning of basic robotic and automated systems.				
CO4		To analyze current applications and emerging trends in AI and robotics, along with their ethical and societal implications.				
Course Description		This course provides an introductory overview of Artificial Intelligence and automation, covering fundamental concepts, techniques, and real-world applications. Students will learn about intelligent agents, basic problem-solving methods, and emerging AI trends. The course aims to build a strong foundation for further studies in intelligent systems and automation technologies.				
Detailed Syllabus						
Unit	Topics				CO No.	Number of lectures
Unit 1: Introduction Artificial Intelligence	Definition and scope of Artificial Intelligence (AI), Historical context and evolution of AI-Turing Test and Chinese room argument, Problem reduction State space search: Problem-solving algorithm, Major Components of AI and Search Process				CO1	6

Unit 2: Intelligent Agents and Problem Solving	Introduction to Agents and environment, nature of environment and agents, problem solving agent, Search Algorithm: Brute Force Search, Depth First Search, Breadth First Search, Heuristic Search Algorithm: Hill Climbing Algorithm, Best First Search, A and AO* Algorithm, Knowledge Representation	CO2	7
Unit 3: Introduction to Robotics and Automation	Introduction to robotics, Basic Concepts, History and Classification, Types of Robots and Automation Systems like Industrial robots, mobile robots, humanoid robots, Smart appliances, Key Components: Sensors, Actuators, Controllers, Introduction to Control Systems, Automation in Manufacturing, Smart Homes, and Agriculture	CO2, CO3	9
Unit 4: Applications of Robotics and Future Trends	Aerial Robotics: Introduction to Drones (UAVs), Components: Frame, motors, propellers, flight controller, Flight principles: Lift, thrust, drag, Applications: Surveillance, mapping, delivery. Emerging Trends in Robotics: Swarm Robotics, Collaborative Robots (Cobots), Soft Robotics, Human-Robot Interaction (HRI)	CO3, CO4	9
Text Books :			
1. Machine Learning, Tom Mitchell, McGraw Hill, 1997. 2. S. Russell and P. Norvig, "Artificial Intelligence: A Modern Approach", Prentice Hall, Third Edition, 2009. 3. "Introduction to Robotics: Mechanics and Control" by John J. Craig			
Reference Books :			
1. Bratko, I. (2011). <i>Prolog: Programming for Artificial Intelligence</i> (4th ed.). Addison-Wesley Educational Publishers Inc. 2. Janakiraman, V. T., Sarukesi, K., & Gopalakrishnan, P. (2001). <i>Foundations of Artificial Intelligence and Expert Systems</i> . New Delhi: Macmillan India Ltd. 3. Introduction to AI Robotics" by Robin R. Murphy 4. Drone Fundamentals: Theory and Practice for Drone Pilots by David G. Price			
Online Resources:			
1. https://pll.harvard.edu/course/cs50s-introduction-artificial-intelligence-python 2. https://ocw.mit.edu/			
Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination			
Examination Scheme:			
Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
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CO2	-	2	1	-	2	-	-	-	-	-	-	-	-	-
CO3	-	2	1	-	2	-	-	-	-	-	-	-	-	-
CO4	-	2	1	-	2	-	-	-	-	-	-	-	-	-
Average	-	2	1	-	2	-	-	-	-	-	-	-	-	-

IILM University, Greater Noida	School:	School of Computer Science & Engineering	Year:	I	Course Name:	<i>Digital Electronics and Computer Organisation Lab</i>
					Course Code:	CSE26102P
	Program:	B. Tech. (CSE)	Semester:	I	L-T-P-C	0-0-2-1
					Prerequisite Subject:	Digital Electronics
Course Objectives	1. To provide hands-on experience with basic digital components and logic gates					
	2. To develop practical skills in using simulation tools and digital kits for implementing and testing digital circuits and verifying theoretical concepts in real-time.					
	3. To reinforce understanding of computer organization concepts, including register transfer, memory interfacing, and basic CPU operations through simulation and small-scale hardware experiments.					
	4. To cultivate the ability to design, simulate, and debug digital systems and processor-based modules, preparing students for advanced courses in embedded systems, microprocessors, and computer architecture.					
Course Outcome	Description					

(At the end of the course, students will be able to)	
CO1	Understand the basic principles of Boolean algebra using logic gates and K-Map technique.
CO2	Analyse and design combinational and sequential logic circuits.
CO3	Understand the functional units of a computer system including the addressing modes and processor performance.
CO4	Apply the approaches aimed at achieving communication between processor and I/O devices.
CO5	Analyse internal memory organization and importance of Pipelining on processor performance.
Course Description	The Digital Electronics and Computer Organisation Lab is designed to provide hands-on experience in fundamental digital electronics concepts and their practical applications. Understand the basic structure of a computer, including buses, ALU operations, and instruction execution. This lab bridges theoretical knowledge with practical implementation, equipping students with the skills to design, analyse, and optimize digital systems for real-world applications.
List Of Experiments	1. Introduction to Digital Electronics and Logic Gates To understand the nomenclature, specifications, and functioning of basic digital ICs and verify the truth tables of logic gates. 2. Implementation of Boolean Functions using Logic Gates To implement Boolean expressions using basic gates in SOP and POS forms. 3. Study of Half Adder and Full Adder To understand and verify the operation of half adder and full adder circuits. 4. Study of Half Subtractor and Full Subtractor To understand and implement subtraction operations using logic gates. 5. Study of Multiplexer and Demultiplexer To explore the working of multiplexers and demultiplexers. 6. Decoder/De-Multiplexer and Encoder Implementation To implement and verify the functionality of encoders and decoders using logic gates. 7. Verification of RS and JK Flip-Flops To study the operation of RS and JK flip-flops. 8. Study of Synchronous Counter To understand and implement a synchronous counter circuit. 9. Simulation of Simple ALU Operations To simulate basic arithmetic and logical operations using digital components. 10. Design of Memory Unit in Multisim To design and simulate a basic memory unit with read/write capabilities. 11. Register Transfer Logic Implementation To implement and simulate basic operations using register transfer logic.

Experiment 1:	Title: Introduction to Digital Electronics and Logic Gates Objective: To understand the nomenclature, specifications, and functioning of basic digital ICs and verify the truth tables of logic gates. Lab Activities: 1. Verify truth tables of NOT, AND, OR, NAND, NOR, XOR, and XNOR gates using ICs.
Experiment 2:	Title: Implementation of Boolean Functions using Logic Gates Objective: To implement Boolean expressions using basic gates in SOP and POS forms. Lab Activities: 1. Design and implement given Boolean functions using logic gates. 2. Verify the outputs for SOP and POS implementations.
Experiment 3:	Title: Study of Half Adder and Full Adder Objective: To understand and verify the operation of half adder and full adder circuits. Lab Activities: 1. Design and implement half adder and full adder circuits using basic gates. 2. Verify their truth tables using input combinations
Experiment 4:	Title: Study of Half Subtractor and Full Subtractor Objective: To understand and implement subtraction operations using logic gates. Lab Activities: 1. Design and implement half subtractor and full subtractor circuits. 2. Verify outputs for all input conditions
Experiment 5:	Title: Study of Multiplexer and Demultiplexer Objective: To explore the working of multiplexers and demultiplexers. Lab Activities: 1. Implement basic MUX and DEMUX circuits. 2. Demonstrate data selection and distribution using control inputs.
Experiment 6:	Title: Decoder/De-Multiplexer and Encoder Implementation Objective: To implement and verify the functionality of encoders and decoders using logic gates. Lab Activities: 1. Design encoder and decoder circuits. 2. Verify outputs with various input combinations.
Experiment 7:	Title: Verification of RS and JK Flip-Flops Objective: To study the operation of RS and JK flip-flops. Lab Activities: 1. Construct RS and JK flip-flops using logic gates or ICs.

	2. Verify the characteristic table by applying input combinations
Experiment 8:	Title: Study of Synchronous Counter Objective: To understand and implement a synchronous counter circuit. Lab Activities: 1. Design a binary synchronous counter using flip-flops. 2. Verify its operation by monitoring the output sequence.
Experiment 9:	Title: Simulation of Simple ALU Operations Objective: To simulate basic arithmetic and logical operations using digital components. Lab Activities: 1. Simulate ALU operations like addition, subtraction, AND, OR, etc. 2. Validate the functionality using sample inputs in simulation tools.
Experiment 10:	Title: Design of Memory Unit in Multisim Objective: To design and simulate a basic memory unit with read/write capabilities. Lab Activities: 1. Create a simple RAM circuit in Multisim. 2. Demonstrate read and write operations.
Experiment 11:	Title: Register Transfer Logic Implementation Objective: To implement and simulate basic operations using register transfer logic. Lab Activities: 1. Design RTL circuits for data movement and arithmetic operations. 2. Simulate and verify the operation of each instruction.
Text Books:	
1. "Digital Logic and Computer Design" by M. Morris Mano 2. "Computer Systems: A Programmer's Perspective" by Randal E. Bryant and David R. O' Hallaron	
Reference Books :	
1. Carl Hamacher, Zvonks Vranesic, SafeaZaky (2002), Computer Organization, 5th edition, McGraw Hill, New Delhi, India. 2. William Stallings (2010), Computer Organization and Architecture- designing for performance, 8th edition, Prentice Hall, New Jersey. 3. Anrew S. Tanenbaum (2006), Structured Computer Organization, 5th edition, Pearson Education Inc. 4. John P. Hayes (1998), Computer Architecture and Organization, 3rd edition, Tata McGrawHill	
Online Resources:	

1. NPTEL / SWAYAM –offered by IIT Kharagpur “Computer Architecture and Organization”
https://onlinecourses.nptel.ac.in/noc21_cs61/preview
2. “Computer Organization & Architecture” Playlist by Engineering Funda
<https://www.youtube.com/playlist?list=PLgwJf8NK-2e7XZXcFujMw--IDZ2nnsXNT>

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination
Examination Scheme:

Components	Continuous Assessment	Lab Test / Presentation / Assignment / etc.
Weightage (%)	50%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO 11	PO12	PSO1	PSO2
CO1	3	2	1	-	-	-	-	-	-	-	-	2	3	-
CO2	3	3	2	-	2	-	-	-	1	-	-	2	3	2
CO3	3	3	3	3	2	2	-	-	-	2	-	3	3	3
CO4	3	2	2	2	2	2	-	-	-	-	-	2	2	3
CO5	3	2	2	2	1	3	-	-	-	-	-	2	2	3
Average	3	2.4	2	1.4	1.4	1.4	-	-	0.2	0.4	-	4.6	2.6	2.2

IILM University, Greater Noida	School	School of Computer Science & Engineering	Year:	I	Course Name	Venture Ideation
					Course Code	
	Program :	B.Tech. (CSE)	Semester :	II	L-T-P-C	0-0-4-2
					Prerequisite Subject	-
Course Objectives		1.To introduce the concepts of innovation, creativity, and entrepreneurship in the context of emerging technologies.				
		2. To empower students to identify real-world challenges and generate impactful venture ideas.				
		1. To facilitate experiential learning through ideation, validation, prototyping, and pitching.				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		To identify societal, industrial, or consumer problems and link them to entrepreneurial opportunities.				
CO2		To apply ideation and design thinking tools for developing innovative ideas.				
CO3		To validate ideas through stakeholder interviews and market research.				
CO4		To develop a business model canvas and present a venture pitch effectively.				

Course Description		The course introduces B.Tech students to the fundamentals of innovation and entrepreneurship, emphasizing venture ideation through design thinking. It fosters a mindset of creative problem solving, opportunity recognition, and the translation of ideas into validated concepts. Students will learn the structured process of ideating, validating, and presenting a technology-based venture through experiential learning and hands-on activities.													
Detailed Syllabus															
Unit	Topics										CO No.	Number of lectures			
Unit 1: Introduction to Innovation and Entrepreneurship	Concept and types of innovation (sustaining, disruptive, incremental), Introduction to entrepreneurship and entrepreneurial mindset, Key traits of successful entrepreneurs, Innovation ecosystems: Government policies, Atal Innovation Mission, Startup India.										CO1	5			
Unit 2: Problem Identification and Idea Generation	Problem discovery and framing, Opportunity identification Tools for idea generation: Brainstorming, SCAMPER, TRIZ, lateral thinking, Case studies on student-led/start-up innovations.										CO2	7			
Unit 3: Design Thinking and Validation	Introduction to design thinking (Empathize, Define, Ideate, Prototype, Test), Empathy maps, user research, and persona building, Customer discovery, problem-solution fit, MVP (Minimum Viable Product) and prototyping basics.										CO2, CO3	9			
Unit 4: Business Model and Pitching	Business Model Canvas (BMC) – nine building blocks, Value Proposition Canvas, Basics of startup finance and revenue models, Pitch deck elements and elevator pitch preparation										CO3, CO4	9			
Text Books :															
1. Samson, D., & Gloet, M. (2016). <i>Innovation and entrepreneurship: Creating new value</i> (ISBN 978-0190300630). Oxford University Press.															
Reference Books :															
1. AICTE Innovation Cell Handbook (available at: https://mic.gov.in) Kelley, T., & Kelley, D. (2013). <i>Creative confidence: Unleashing the creative potential within us all</i> . Crown Business.															
Online Resources:															
1.Innovation and Entrepreneurship: measiim.edu.in/myweb/uploads/2022/05/PMFEA-IE-1.pdf 2. Niti Ayog: niti.gov.in/sites/default/files/2021-09/Innovation-andEntrepreneurship-forSustainableGrowth_0.pdf															
Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination															
Examination Scheme:															
Components		MSE			Presentation/Assignment/ etc					ESE					
Weightage (%)		20%			30%					50%					
Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)															
CO	PO 1	P O 2	P O 3	P O4	PO 5	P O 6	P O 7	P O 8	P O 9	P O 10	P O 11	P O 12	P S O 1	P S O 2	

CO1	2	3	1	1	2	2	1	1	3	2	1	2	-	-
CO2	1	3	1	2	2	1	2	1	2	2	2	2	-	-
CO3	2	3	1	2	2	1	2	1	2	2	2	3	-	-
CO4	1	2	1	1	2	1	2	1	3	3	3	3	-	-
Average	1.5	2.75	1	1.5	2	1.2	1.75	1	2.5	2.5	2	2.5	-	-

IILM University, Greater Noida	School	School of Computer Science & Engineering	Year:	I	Course Name	Venture Ideation
					Course Code	CSE26109
	Program :	B.Tech. (CSE)	Semester :	I	L-T-P-C	0-0-4-2
					Prerequisite Subject	-
Course Objectives		1.To introduce the concepts of innovation, creativity, and entrepreneurship in the context of emerging technologies.				
		2. To empower students to identify real-world challenges and generate impactful venture ideas.				
		1. To facilitate experiential learning through ideation, validation, prototyping, and pitching.				
Course Outcome (At the end of the course, students will be able to)		Description				
CO1		To identify societal, industrial, or consumer problems and link them to entrepreneurial opportunities.				
CO2		To apply ideation and design thinking tools for developing innovative ideas.				
CO3		To validate ideas through stakeholder interviews and market research.				
CO4		To develop a business model canvas and present a venture pitch effectively.				
Course Description		The course introduces B.Tech students to the fundamentals of innovation and entrepreneurship, emphasizing venture ideation through design thinking. It fosters a mindset of creative problem solving, opportunity recognition, and the translation of ideas into validated concepts. Students will learn the structured process of ideating, validating, and presenting a technology-based venture through experiential learning and hands-on activities.				
Detailed Syllabus						
Unit	Topics				CO No.	Number of lectures
Unit 1: Introduction to Innovation and Entrepreneurship	Concept and types of innovation (sustaining, disruptive, incremental), Introduction to entrepreneurship and entrepreneurial mindset, Key traits of successful entrepreneurs, Innovation ecosystems: Government policies, Atal Innovation Mission, Startup India.				CO1	5
Unit 2: Problem Identification and Idea Generation	Problem discovery and framing, Opportunity identification Tools for idea generation: Brainstorming, SCAMPER, TRIZ, lateral thinking, Case studies on student-led/start-up innovations.				CO2	7

Unit 3: Design Thinking and Validation	Introduction to design thinking (Empathize, Define, Ideate, Prototype, Test), Empathy maps, user research, and persona building, Customer discovery, problem-solution fit, MVP (Minimum Viable Product) and prototyping basics.	CO2, CO3	9
Unit 4: Business Model and Pitching	Business Model Canvas (BMC) – nine building blocks, Value Proposition Canvas, Basics of startup finance and revenue models, Pitch deck elements and elevator pitch preparation	CO3, CO4	9

Text Books :

1. Samson, D., & Gloet, M. (2016). *Innovation and entrepreneurship: Creating new value* (ISBN 978-0190300630). Oxford University Press.

Reference Books :

1. AICTE Innovation Cell Handbook (available at: <https://mic.gov.in>)

Kelley, T., & Kelley, D. (2013). *Creative confidence: Unleashing the creative potential within us all*. Crown Business.

Online Resources:

1. Innovation and Entrepreneurship: measiim.edu.in/myweb/uploads/2022/05/PMFEA-IE-1.pdf

2. Niti Ayog: niti.gov.in/sites/default/files/2021-09/Innovation-andEntrepreneurship-forSustainableGrowth_0.pdf

Modes of Evaluation: Quiz/Assignment/ Presentation/ Extempore/ Written Examination

Examination Scheme:

Components	MSE	Presentation/Assignment/ etc	ESE
Weightage (%)	20%	30%	50%

Relationship between the Course Outcomes (COs), Program Outcomes (POs) and Program Specific Outcomes (PSOs)

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2
CO1	2	3	1	1	2	2	1	1	3	2	1	2	-	-
CO2	1	3	1	2	2	1	2	1	2	2	2	2	-	-
CO3	2	3	1	2	2	1	2	1	2	2	2	3	-	-
CO4	1	2	1	1	2	1	2	1	3	3	3	3	-	-
Average	1.5	2.75	1	1.5	2	1.2	1.75	1	2.5	2.5	2	2.5	-	-